ONLINE STORE · HYPOTHESIS PRIORITIZATION & A/B TEST

Group B converts better. Roll it out.

Which marketing idea to test first, and whether a running A/B test should be stopped

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PRIORITIZATION

Adding reach changes what goes first

Hypothesis	ICE	RICE	Rank change
Subscription form on all main pages	11.2 (3rd)	**112 (1st)**	up 2
Product recommendation blocks	7.0 (5th)	56 (2nd)	up 3
Two new traffic channels	13.3 (2nd)	40 (3rd, tie)	down 1
Banners with current offers	8.0 (4th)	40 (3rd, tie)	up 1
Discount promotion for users	16.2 (1st)	16.2 (5th)	down 4

The discount promotion scores highest on impact and confidence, but it reaches few users. The subscription form reaches almost everyone, so it should be tested first.

Top 5 of 9 hypotheses. ICE = Impact × Confidence / Effort · RICE = Reach × Impact × Confidence / Effort

DATA CHECK

Before testing anything: 58 users were in both groups

58
users in A and B

181
orders removed

1,016
orders analysed
(from 1,197)

A user who sees both versions breaks the independence between groups and can push both metrics at once. They were removed before any metric was calculated, so the comparison uses clean, independent samples.

CONVERSION

Group B converts about 16% to 17% better, and it holds

RAW DATA



+16.0% for B · p = 0.011 · **significant**

WITHOUT OUTLIERS



+17.4% for B · p = 0.016 · **significant**

Removing unusual users and orders does not erase the gap. B's advantage comes from more visitors placing an order, not from a few extreme buyers.

Mann-Whitney U test, α = 0.05. Outliers: users with more than 1 order or an order above the 95th percentile (414.28)

AVERAGE ORDER SIZE

The order value gap was one huge order, not a real effect

RAW DATA



+27.8% for B · p = 0.862 · not significant

WITHOUT OUTLIERS



−3.3% for B · p = 0.771 · not significant

A single order of about 19,920 in group B made its cumulative revenue jump on 19 August. Once outliers are removed, the difference disappears and even reverses.

Mann-Whitney U test, α = 0.05

Recommendation

Stop the test

Declare group B the winner on conversion. The result is significant on raw and filtered data.

Don't sell it as revenue per order

There is no evidence that B raises order value. The gain comes from more people buying.

Next test

Start with the subscription form on all main pages, the top RICE priority with the widest reach.

Clean the groups first, check outliers before trusting a jump, and keep only the conclusions that survive both.

Notebook: github.com/lucianajost/data_analytics_portfolio